Equations with brackets

1 Use substitution to determine whether the given value of x is the solution to each equation:

a $x = 6$ where $2(x - 3) = -8$.

b $x = 8$ where $7(x - 6) = 14$.

c $x = 8$ where $3(x - 6) = 6$.

d $x = 8$ where $9 = 7(x - 7)$.

2 Solve the following equations by first expanding the brackets:

a $2(x + 2) = 0$

b $7(x - 5) = 0$

c $4(x + 5) = 32$

d $5(x + 5) = 39$

e $4(x + 6) = -60$

f $12(x + 5) = 168$

g $3(x - 4) = -21$

h $4(x - 6) = -5$

i $5(x - 4) = -15$

j $12(x - 6) = 12$

k $5(2x + 6) = 69$

l $6(3x + 5) = 192$

m $9(3x + 5) = 153$

n $6(2x - 6) = -90$

o $6(3x - 5) = -102$

p $9(3x - 5) = 63$

q $-6(4x + 5) = -143$

r $-6(3x + 6) = 54$

s $-8(4x + 5) = 120$

t $-6(2x - 4) = 81$

u $-6(4x - 5) = 174$

v $-8(2x - 6) = -16$

3 Solve the following equations:

a $4(2x + 5) + 4 = 48$

b $6(4x + 8) - 9 = 87$

c $5(4x + 5) + 3x = 71$

d $4(2x - 6) - 3x = 16$

e $5(2x + 5) + 4x + 6 = 59$

f $4(4x - 6) - 3x + 8 = 36$

g $-3(4x + 7) + 3x - 5 = -8$

h $-4(4x + 7) - 3x + 5 = 15$

i $-3(4x + 6) - 3x - 7 = -70$

j $-4(4x + 6) + 3x + 7 = -56$

4 Solve the following equations:

a $4(3x + 5) + 3(2x + 6) = 74$

b $2(2x - 5) + 3(4x + 6) = 56$

c $2(4x + 5) + 3(3x - 6) = -42$

d $3(4x + 6) - 2(3x + 5) = 26$

e $5(4x - 6) - 3(2x + 5) = -73$

f $8(4x + 5) - 3(2x - 6) = -46$

g $-3(2x + 5) + 5(4x + 6) = 57$

h $-3(2x + 5) + 5(3x + 6) = 42$

i $-3(2x + 6) + 5(4x - 5) = -15$

j $-3(2x - 4) + 5(3x + 5) = 55$

k $-3(2x - 6) + 5(4x - 5) = 21$

l $-3(2x + 4) + 5(3x + 5) = 40$

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Applications

5 A square has a side length of $4x + 5$ cm. If the perimeter of the square is 44 cm, find the value of x.

6 A rectangle has a width of $5x - 3$ cm and a height of $3x + 7$ cm. If the perimeter of the rectangle is 42 cm, find the value of x.

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